STUDY / PRESENTATION GUIDE

Steganographic Signing and Filter Traceability

- A clear route through the robustness benchmark, its evidence, and its limits
- MSc Computer Science project | Opaleye Toluwalope Abayomi | 24163800
- Evidence baseline: validated final1200 evidence set
- Core question: Which verification signals remain useful after controlled image transformations?
- Reading rule: every result is conditional on the implementation, dataset, transform, metric, and validation state. A recovered signal is not proof of truth, identity, or provenance.

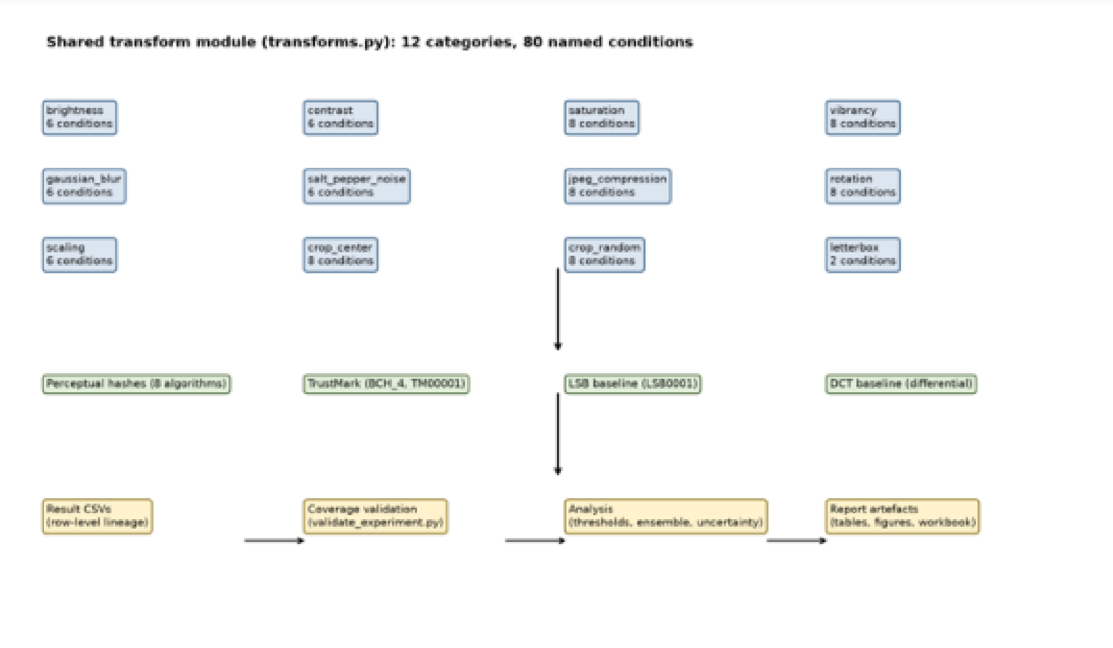

Figure 1. Transform taxonomy used to organise the robustness study.

1. EXECUTIVE OVERVIEW

The central message

- This project benchmarks complementary image-verification signals rather than searching for one universal authenticity detector.
- TrustMark is generally strong against value and encoding changes in the tested evidence, while rotation, cropping, and letterbox padding expose important weaknesses.
- Perceptual hashes provide a separate similarity signal. LSB and DCT provide useful baselines but are not production recommendations.
- The ensemble is a decision aid, not a trust system.
- Suggested opening: Images are routinely resized, recompressed, filtered, cropped, and re-encoded. This project asks whether several verification signals degrade gracefully under those operations.

2. STUDY DESIGN

A shared, deterministic experiment

- Dataset: MS-COCO 2017 validation images, selected deterministically by filename order.
- Evidence scale: validated final1200 run; image is the statistical unit for uncertainty summaries.
- Transform matrix: 12 transform families and 80 transform-intensity conditions.
- Hash layer: eight perceptual-hash variants; reference-to-transformed Hamming distance.
- Watermark layer: TrustMark, LSB, and differential DCT baselines with fixed payloads.
- Analysis: threshold crossings, a worst-algorithm hash rule, and best/fallback ensemble choices.
- Transform families: photometric; encoding/noise; geometric; and black or grey letterbox layout changes.

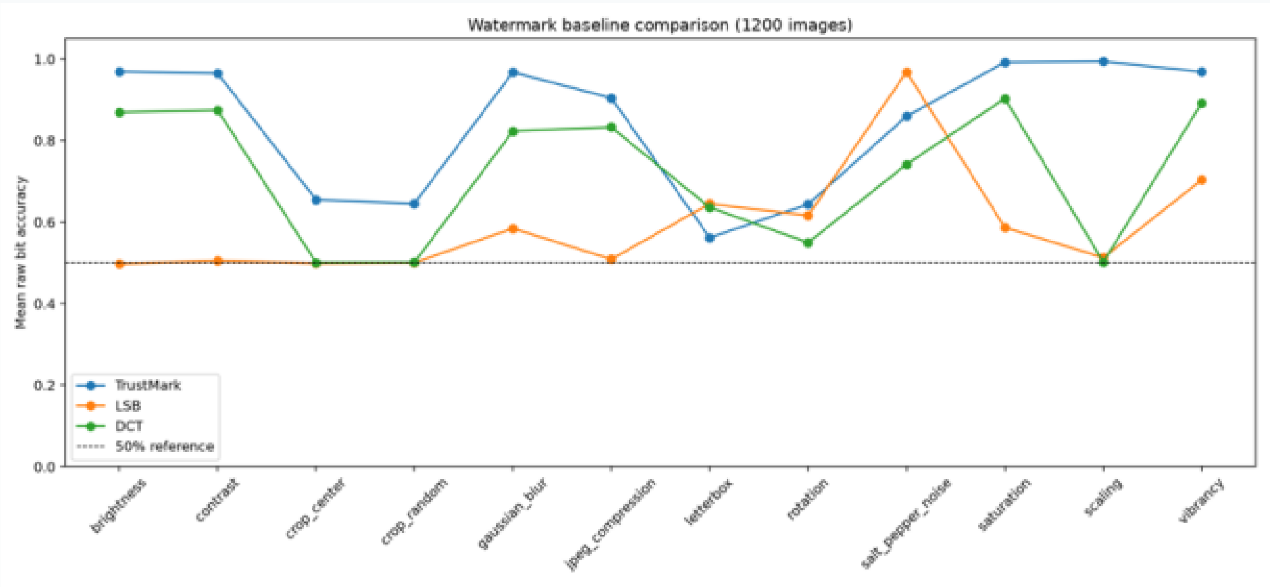

Figure 2. Method-comparison view framing the complementary roles of hashes and watermarking.

3. SIGNALS AND METRICS

What each number means

- Perceptual hashes: Hamming distance counts changed bits; bit-error fraction normalises distance by hash length. PDQ is represented as 256 bits.
- TrustMark uses a learned encoder and decoder. LSB writes to the red-channel least-significant bit plane. DCT uses differential mid-frequency coefficients.
- Raw bit accuracy is recovered payload bits before error correction. Decode presence is whether the implementation reports a valid decoded payload.
- PSNR and MSE describe encode quality, not robustness or security.
- Watermark success is mean bit accuracy above 0.5. Hash success is mean bit-error fraction below 0.5.
- The 0.5 boundary is an analysis convention, not a security proof or user-validated operating point.

4. FINDINGS TO PRESENT

Geometry exposes the failure surface

- Brightness and contrast: LSB crosses the 50% boundary at selected settings; hash and TrustMark remain below their critical thresholds in the reported table.
- JPEG: LSB crosses at quality 20; DCT remains the more informative classical comparator in this study.
- Rotation: DCT crosses at 1 degree; the worst-hash rule at 70 degrees; LSB at 90 degrees.
- Scaling: LSB crosses at 0.25x; DCT crosses at 0.25x, with a non-monotonic worst crossing at 3.0x.
- Centre and random crop: worst-hash crossings occur at 0.7 and 0.6 removal respectively; baselines cross earlier.
- Letterbox padding changes layout without removing source content, making it a useful condition-specific test.

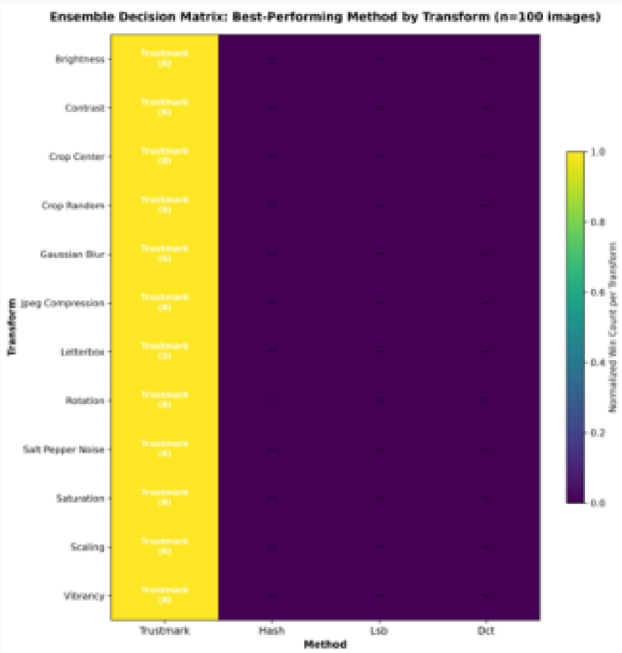

Figure 3. Ensemble heatmap showing method choices across transform-intensity conditions.

5. EVIDENCE WALKTHROUGH

Recommended presentation sequence

- 1. Problem: show the transform taxonomy and explain routine edits versus geometric disruption.
- 2. Design: show the shared pipeline and four signal families.
- 3. Metrics: explain Hamming distance, raw bit accuracy, decode presence, and PSNR.
- 4. Results: use the comparison and heatmap to contrast failure regions.
- 5. Dashboard: open overview, per-transform, ensemble, and per-image views.
- 6. Boundary: state what the results do not establish.
- The dashboard is a CSV-backed research visualisation and evidence browser, not a production signing service.

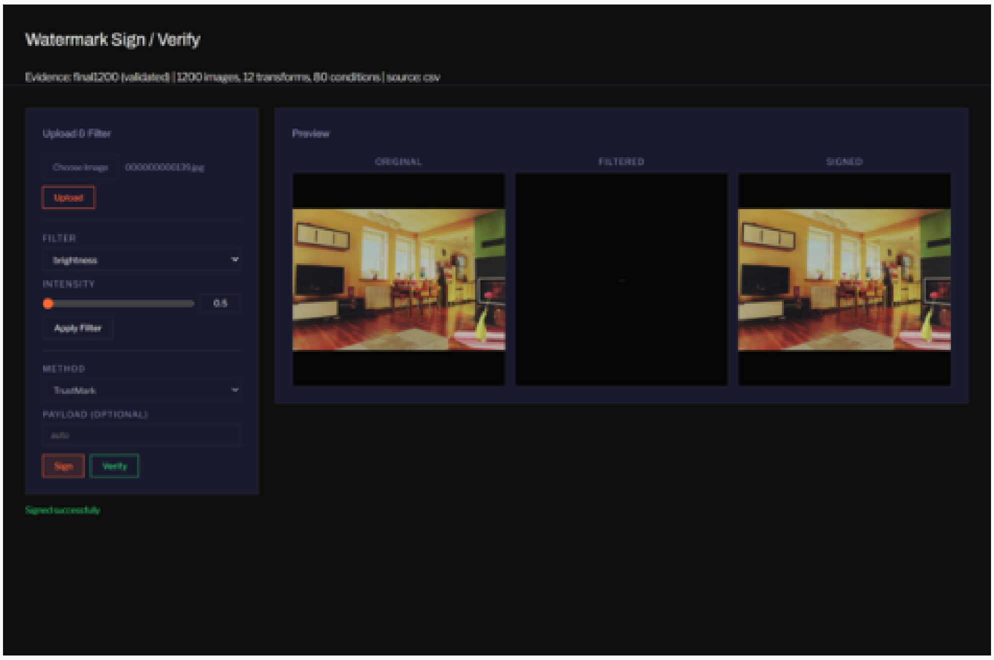

Figure 4. Dashboard interface for navigating the benchmark evidence.

6. QUALITY AND REPRODUCIBILITY

Controls that support trust in the measurements

- Transform definitions are centralised in transforms.py; image selection and random seeds are deterministic.
- Validation checks coverage, expected rows, duplicate keys, missing values, and baseline records before interpretation.
- Raw CSV outputs are retained separately from summaries and dashboard views.
- Uncertainty summaries use image-level aggregation, fixed-seed percentile bootstrap intervals for continuous metrics, and Wilson intervals for decode proportions.
- Corrections retained in the evidence: PDQ serialisation uses one bit per dimension; TrustMark records raw decoder output separately from exact payload presence.
- A configured larger run is not completed evidence until all pipelines pass validation.

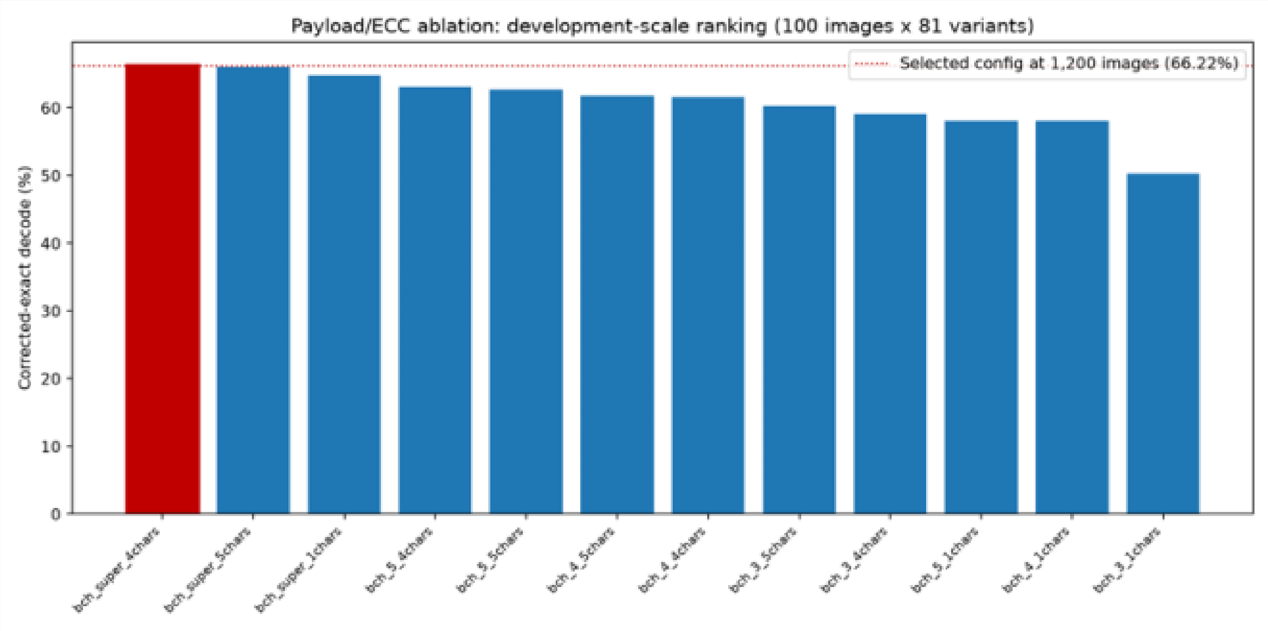

Figure 5. Payload and error-correction development sweep used to motivate configuration choices.

7. RESPONSIBLE USE

Limitations and safe wording

- The evidence is tied to selected MS-COCO images and does not establish performance on medical, synthetic, text-heavy, or customer imagery.
- Transformations are isolated; compound, adaptive, and adversarial attacks are outside the demonstrated scope.
- The ensemble is comparative thresholded evidence, not calibrated authentication accuracy or decision cost.
- C2PA manifests, certificates, signed claims, key management, identity, and a trust chain are not implemented.
- Public images may contain recognisable people or copyrighted material. Future payloads containing identifiers require purpose, lawful basis, retention, access, and deletion controls.
- Prefer: ‘useful signal under the tested condition’. Avoid: ‘proves the image is authentic’.

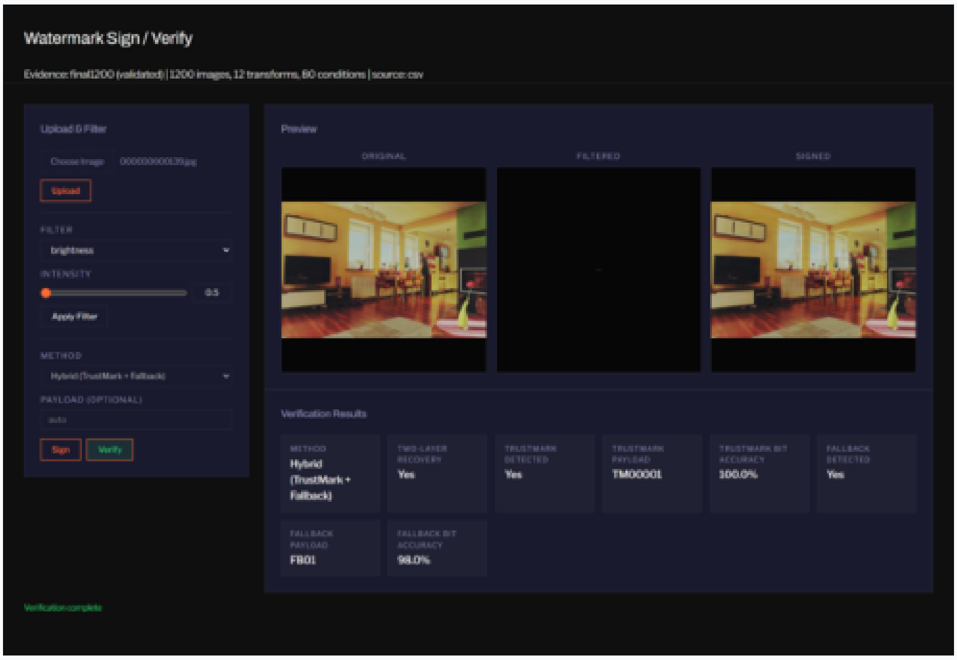

Figure 6. Hybrid verification view illustrating complementary signal inspection.

8. QUESTIONS AND SHORT ANSWERS

Defensible answers for discussion

- Why not only a watermark? A watermark and a hash measure different relationships; a hybrid can preserve evidence when one signal degrades.
- Why is geometry difficult? Rotation, crop, scaling, and padding alter coordinate relationships.
- Does high bit accuracy mean security? No. It reports recovery under tested conditions; security also needs a threat model, key management, and attack evaluation.
- Why report raw accuracy and decode presence? Error correction can make exact decoding succeed even when raw bits are wrong.
- Why use the worst hash? A pooled mean can hide a weak algorithm; the weakest-link rule makes that choice explicit.
- What next? Complete validated larger-run evidence, add compound and adaptive attacks, report per-image uncertainty and error costs, and evaluate provenance controls separately.

APPENDIX A. SOURCE MAP

Repository evidence used

- Project scope and pipeline components: PROJECT_SUMMARY.md; README.md
- Method definitions and limitations: output/final_project_report_draft.md
- Final-run uncertainty method: output/results/final1200/uncertainty_summary.md
- Threshold crossings: output/results/final1200/threshold_analysis.md
- Figures: output/figures/*.png; each included figure has an asset caption.
- Evidence and submission context: SUBMISSION_MANIFEST.md; CITATIONS.md
- Status: this is a study and presentation document derived from repository evidence. It is not a replacement for the formal project report or a signed provenance implementation.

APPENDIX B. GLOSSARY (1/2)

Terms as used in this guide

- Authenticity — claim that content is true/attributed; not proven by recovery.
- BCH — error-correcting code in watermark packets.
- Bit accuracy — recovered bits before correction.
- Decode presence — valid payload reported; distinct from raw accuracy.
- DCT — frequency-domain baseline in mid-frequency coefficients.
- ECC — converts partial recovery into valid messages.
- Ensemble — per-condition best/fallback matrix.
- False acceptance — accepting unrelated image as verified.
- Hamming distance — differing bits between hashes.
- Integrity — unchanged bytes; needs hard binding.
- Letterbox — padding for aspect ratio, separate from crop.

APPENDIX B. GLOSSARY (cont.) (2/2)

Terms as used (continued)

- LSB — fragile spatial-domain baseline.
- MSE — pixel distortion measure.
- Payload — fixed test message, not an identifier.
- Perceptual hash — content-derived representation vs distance.
- pHash/dHash — hash variants among eight tested.
- Provenance — signed accountable record; needs manifests/trust chain.
- PSNR — imperceptibility metric, not robustness.
- Recovery — payload decodes (decode presence).
- TrustMark — learned neural watermark candidate.
- Two-layer fallback — second BCH DCT layer; OR recovery.
- Transform intensity — named condition param, only within-family comparable.